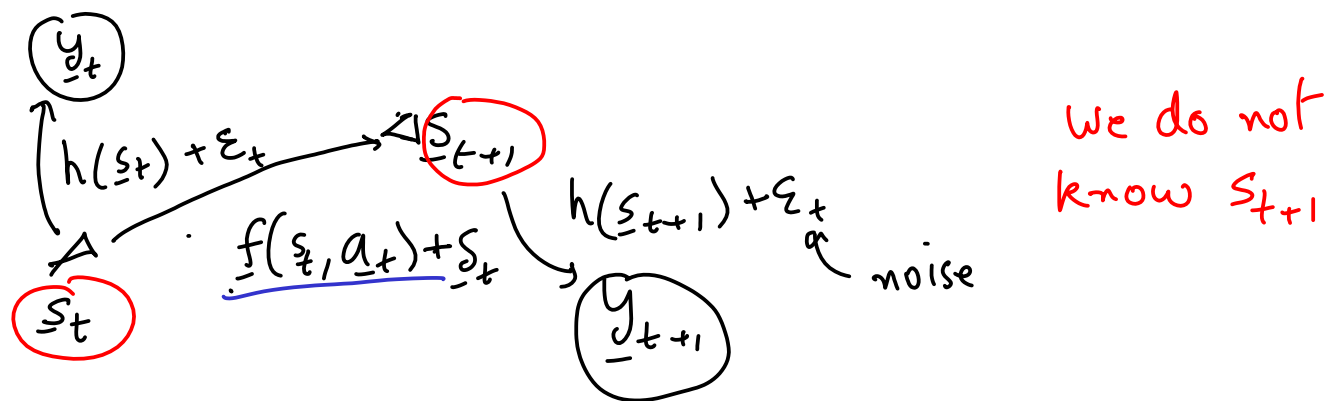


Localization and Mapping

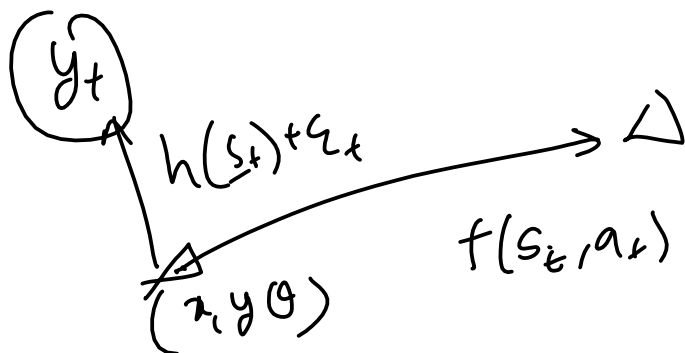
State estimation



Using the observations, find s_{t+1}

If we know s_t and $f(s_t, a_t)$, then
 $s_{t+1} = f(s_t, a_t) + \delta_t$
 (where δ_t is noise)

State estimation is the problem of estimating state from observations (noisy) and noisy motion model.



ARUCO markers can give you full state estimate in single observation.
 Full state observability

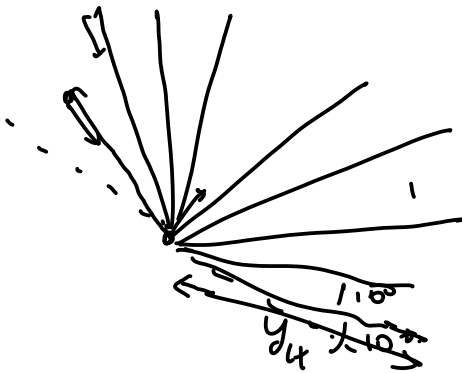
Partial state observability examples :

① Object detection with camera
(given the angle of an object w.r.t the camera but not the distance or the orientation of the object)

② Laser scan will provide angle and distance of surfaces from the laser scan but not their orientation

observation vector $\rightarrow \underline{y}_t = \underset{\substack{\uparrow \\ \text{observation mode}}}{h(\underline{s}_t)} + \underset{\substack{\nwarrow \\ \text{noise}}}{\underline{\varepsilon}_t}$

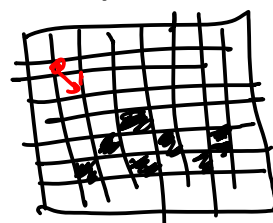
$\underline{\varepsilon}_t \sim \mathcal{N}(\underline{0}_t, \underset{\substack{\nwarrow \\ \text{Covariance matrix}}}{R_t})$
Gaussian Random vector



$$\underline{y}_t = \begin{bmatrix} y_{1t} \\ \vdots \\ y_{nt} \end{bmatrix}$$

$$y_{1t} = \begin{cases} -1 & \text{if it is not reflected} \\ \text{distance from the nearest surface} & \text{if it is reflected} \end{cases}$$

A simple way of representing environment is called occupancy grid. each cell is either occ or free



$$\underline{y}_t = \underline{h}(\underline{s}_t) + \underline{\varepsilon}_t$$

in full observability

$\underline{h}$ is invertible

$$\underline{s}_t \approx \underline{h}^{-1}(\underline{y}_t)$$

in partial observability $\underline{h}$ is not invertible

Noise is stochastic/probabilistic, we use probabilistic tools

Equiv

$$\begin{aligned} & \underline{s}_{t+1} = \underline{f}(\underline{s}_t, a_t) + \underline{\varepsilon}_t, \quad \underline{\varepsilon}_t \sim \mathcal{N}(\underline{0}, Q_t) \\ & \rightarrow P(\underline{s}_{t+1} | \underline{s}_t, a_t) \end{aligned}$$

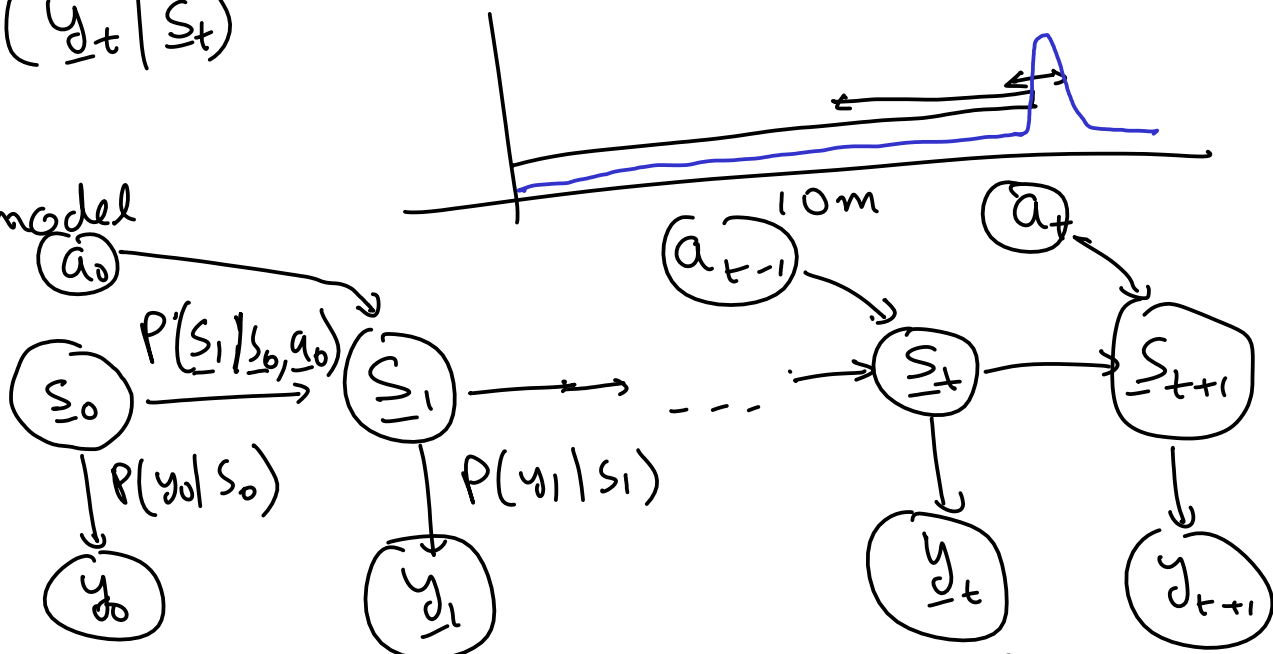
Transition probability, Motion model

Equiv

$$\begin{aligned} & \underline{y}_t = \underline{h}_t(\underline{s}_t) + \underline{\varepsilon}_t, \quad \underline{\varepsilon}_t \sim \mathcal{N}(\underline{0}, R_t) \\ & \rightarrow P(\underline{y}_t | \underline{s}_t) \end{aligned}$$

Prob.

Graphical model



each node is a Random Variable, each edge is a Prob. dist.

Markov Decision Process

$$(S, A, P(s_{t+1}|s_t, a_t), P(y_t|s_t), c(s_t, a_t))$$

Main assumption: States are Markovian

$$(s_{t+1} \perp s_{t-1}, s_{t-2}, s_{t-3}, \dots, s_0) | s_t$$

state captures all the historical informations

Formally, we want to estimate

$$P(s_t | y_t, y_{t+1}, \dots, y_0, a_t, a_{t-1}, \dots, a_0)$$

$$P(y_0 | s_0) = \text{observation model}$$

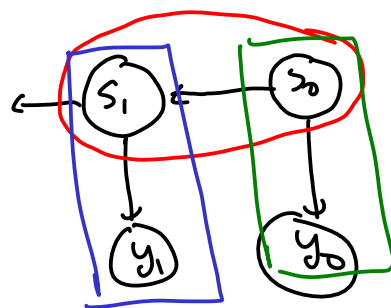
$$P(y_1, y_0 | s_0) = \int_{s_1} P(y_1, y_0, s_1 | s_0) ds_1 \quad \leftarrow \text{Marginalization}$$
$$= \int_{s_1} P(y_1 | s_1, s_0) P(s_1 | s_0) P(y_0 | s_0) ds_1$$
$$= \int_{s_1} \boxed{P(y_1 | s_1)} \boxed{P(s_1 | s_0)} \boxed{P(y_0 | s_0)} ds_1$$

Probabilistic Theorems

① Marginalization

$$P(x) = \sum_y P(x, y) = \int_y P(x, y) dy \quad | \quad P(x|z) = \sum_y P(x, y|z)$$

② conditional independence



$$P(X, Y | Z) = P(X | Z) P(Y | Z) \Leftrightarrow X \perp Y | Z$$

X and Y are mutually independent given Z

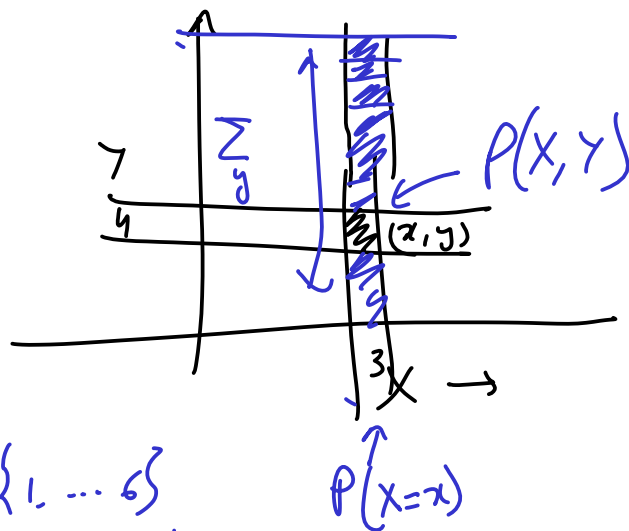
① Marginalization

Joint probability distribution

$$P(X=x, Y=y)$$

$$P(X=x) = \sum_y P(X=x, Y=y)$$

X is rolling a die 6-faces $\{1, \dots, 6\}$
 Y, " " " " $\{1, \dots, 6\}$



② Independence

$$P(X|Y) := \frac{P(X, Y)}{P(Y)}$$

as if Y
 has occurred
 with prob = 1

definition
 of conditional
 prob.

X = learning
 Y = Homework

LHS = $P(X|Y)$ is prob of X given Y

$$P(Y|X) = \frac{P(X, Y)}{P(X)}$$

$$X \perp Y \Leftrightarrow P(X|Y) = P(X) \Leftrightarrow P(Y|X) = P(Y)$$

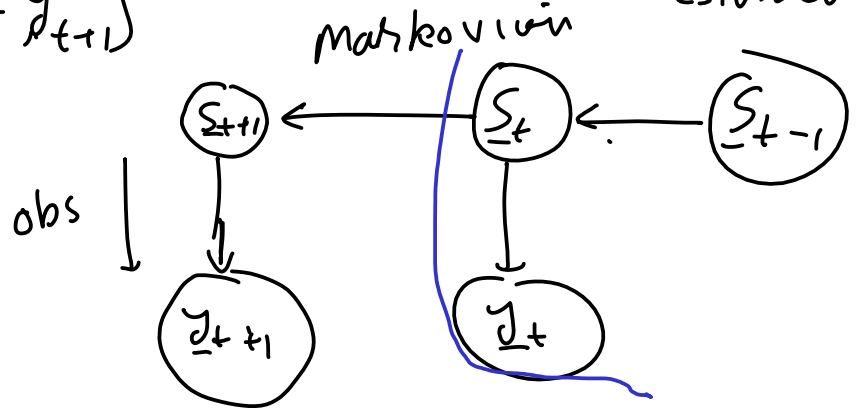
$$\Rightarrow \frac{P(X, Y)}{P(Y)} = P(X) \Rightarrow \boxed{P(X, Y) = P(X) P(Y)} \quad | \quad \text{factorization}$$

$$P(x, y) = P(x|y)P(y) = P(y|x)P(x) \quad \left| \begin{array}{l} \text{Always} \\ \text{true} \end{array} \right.$$

$$P(s_{t+1} | y_1, y_2, \dots, y_{t+1})$$

localization / mapping / state estimation

s_t blocks path
from s_{t-1} to s_{t+1}



$$\left. \begin{array}{l} s_{t+1} \perp s_{t-1} \mid s_t \\ y_{t+1} \perp s_t \mid s_{t+1} \end{array} \right\} \text{Assumptions}$$

$$P(x) = \int_y P(x, y) dy$$

$$P(s_{t+1} | y_1, y_2, \dots, y_{t+1}) = \int_{s_t} P(s_{t+1}, s_t | y_1, y_2, \dots, y_{t+1}) ds_t$$

$$P(x, y | z) = P(x | y, z) P(y | z) = \int_{s_t} P(s_{t+1} | s_t, y_1, \dots, y_{t+1}) \underbrace{P(s_t | y_1, \dots, y_{t+1})}_{ds_t}$$

$$P(x, y) = P(x | y) P(y)$$

$$= \int_{s_t} \underbrace{P(s_{t+1} | s_t, y_1, \dots, y_{t+1})}_{\text{current time step}} \underbrace{P(s_t | y_1, \dots, y_t)}_{\text{prev time step}} ds_{t+1}$$

$$P(\underline{s}_{t+1} | \underline{s}_t, \underbrace{\underline{y}_1, \dots, \underline{y}_{t+1}}_{\underline{y}_t}) = P(\underline{s}_{t+1} | \underline{s}_t, \underline{y}_{t+1})$$

Using $\underline{s}_{t+1} \perp \underline{y}_t | \underline{s}_t$

Bayes Theorem

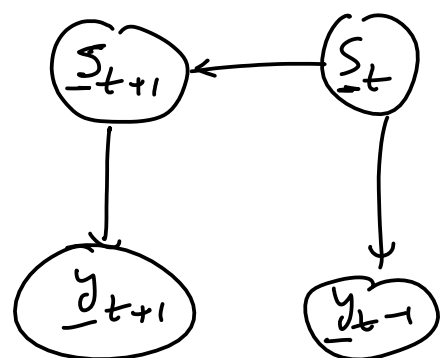
$$P(x, y) = \underbrace{P(x|y)P(y) = P(y|x)P(x)}_{\downarrow}$$

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

$$P(\underline{s}_{t+1} | \underline{s}_t, \underline{y}_{t+1}) = \underbrace{P(\underline{s}_t, \underline{y}_{t+1} | \underline{s}_{t+1})}_{\downarrow} \frac{P(\underline{s}_{t+1})}{P(\underline{s}_t, \underline{y}_{t+1})}$$

$$= \underbrace{P(\underline{s}_t | \underline{s}_{t+1})}_{\downarrow} \underbrace{P(\underline{y}_{t+1} | \underline{s}_{t+1})}_{\checkmark} \frac{P(\underline{s}_{t+1})}{P(\underline{s}_t, \underline{y}_{t+1})}$$



what were the distributions given to us

① Motion model

$$P(\underline{s}_{t+1} | \underline{s}_t, \underline{a}_t) = \underbrace{P(\underline{s}_{t+1} | \underline{s}_t)}$$

$$\underline{s}_t \perp \underline{y}_{t+1} | \underline{s}_{t+1}$$

② Observation model

$$\underbrace{P(\underline{y}_{t+1} | \underline{s}_{t+1})} = P(\underline{y}_t | \underline{s}_t)$$

$$P(\underline{s}_{t+1} | \underline{s}_t, \underline{y}_{t+1}) = P(\underline{s}_{t+1} | \underline{s}_t) P(\underline{s}_t) \frac{1}{P(\underline{s}_t, \underline{y}_{t+1})}$$

$$P(s_{t+1} | s_t, y_{t+1}) = \frac{P(s_{t+1} | s_t) P(y_{t+1} | s_{t+1})}{P(y_{t+1} | s_t)}$$

$\frac{P(s_t, y_{t+1})}{P(s_t)} \rightarrow \frac{P(y_{t+1} | s_t)}{\text{prior}}$

Common
to assume
a uniform prior

$$P(s_{t+1} | y_1, \dots, y_{t+1}) = \int_{s_t} \frac{\overbrace{P(s_{t+1} | s_t)}^{\text{motion model}} \overbrace{P(y_{t+1} | s_{t+1})}^{\text{obs model}}}{P(y_{t+1} | s_t)} P(s_t | y_1, \dots, y_t) ds_t$$

Bayes filter $\rightarrow$ particle filter
 $\quad \quad \quad \hookrightarrow$ Kalman filter